

Problem Set 2

Due Friday, September 25

Instructions. Turn in at least 30 points' worth of problems.

Problem 1. *multiplicative functions and Bell series* (10 points) Let p be a prime. A p -Bell series is a formal Dirichlet series (element of $\text{Dir}(\mathbb{C})$) of the form

$$1 + a_p p^{-s} + a_{p^2} p^{-2s} + a_{p^3} p^{-3s} + \dots = \sum_{k \geq 0} a_{p^k} p^{-ks},$$

where each a_{p^k} is a complex number.

- (a) Show that if $A_p(s)$ is a p -Bell series, then $A_p(s)^{-1}$ exists and is also a p -Bell series.
- (b) Suppose $F_p(s)$ is a p -Bell series for every prime number p . Prove that $\prod_p F_p(s)$ converges in $\text{Dir}(\mathbb{C})$.

Recall that $\text{Dir}(\mathbb{C})$ is a metric space with respect to the "eventual coincide-ence" metric discussed in class: $d(A(s), B(s)) = 2^{-n}$ if n^{-s} is the first place the coefficients of A and B differ, and 0 if $A(s) = B(s)$.

- (c) Prove that the arithmetic function f is multiplicative if and only if $\sum_{n \geq 1} f(n)n^{-s} = \prod_p F_p(s)$ for some choice of p -Bell series $F_p(s)$ (for every prime p).
- (d) Let f, g, h be arithmetic functions, and put $F(s) = \sum_{n \geq 1} f(n)n^{-s}$, $G(s) = \sum_{n \geq 1} g(n)n^{-s}$, and $H(s) = \sum_{n \geq 1} h(n)n^{-s}$. Suppose $F(s)G(s) = H(s)$. Prove that if two of f, g, h are multiplicative, then so is the third.

Problem 2. (10 points) Fix an integer $k \neq 1$. Prove that there are infinitely many composite terms in the sequence $\{2^{2^n} + k\}_{n \geq 1}$.

Hint. First show that the sequence of powers of 2 is eventually periodic mod $p - 1$, for every prime p , with the preperiod length bounded in terms of the exponent of 2 in $p - 1$.

Problem 3. (10 points) For this problem, we let $F_n = 2^{2^n} + 1$ denote the n th Fermat number, for $n = 0, 1, 2, 3, \dots$. Let $x \geq 10$, and let $N = N_x$ be the largest positive integer for which $2^N \leq \sqrt{x}$.

- (a) Show that there is a constant K (independent of x) such that

$$\#\{\text{primes } p : p \mid F_0, F_1, \dots, \text{ or } F_N\} \leq K\sqrt{x}.$$

First, convince yourself that the number of distinct primes dividing m is at most logarithmic in m .

- (b) Show that there is a constant K' (independent of x) such that

$$\#\{\text{primes } p \leq x : p \mid F_n \text{ for some } n > N\} \leq K'\sqrt{x}.$$

Hint. From class, each of the primes counted here is congruent to 1 modulo 2^{N+2} .

- (c) Let $\mathcal{P}$ be the set of primes dividing at least one of $F_0, F_1, F_2, \dots$. Show that $\sum_{p \in \mathcal{P}} \frac{1}{p} < \infty$.

Hint. From (a) and (b), we have that for every $x \geq 10$, the number of elements of $\mathcal{P}$ not exceeding x is at most $K''\sqrt{x}$, for a certain constant K'' . Now estimate the sum over $p \in \mathcal{P}$ by splitting the range of p into intervals $[3, 10)$ together with $[10, 20)$, $[20, 40)$, $[40, 80)$,

Problem 4. (5 points) Show that $2^n + 78557$ is divisible by one of 3, 5, 7, 13, 19, 37, 73 for every positive integer n .

Problem 5. (10 points) Show that there are infinitely many odd numbers k that cannot be written in the form $p + 2^n$, where p is prime and n is a positive integer.

Hint. This condition on k is the same as asking that $k - 2^n$ is never prime.

Problem 6. *p-adic absolute values* (10 points) Every nonzero $x \in \mathbf{Q}$ has a unique decomposition in the form

$$x = \pm \prod_p p^{v_p(x)},$$

where each $v_p(x) \in \mathbf{Z}$ and all but finitely many of the $v_p(x)$ are equal to 0. For each prime p , we set $|x|_p := p^{-v_p(x)}$. Extend $|\cdot|_p$ to all of $\mathbf{Q}$ by setting $|0|_p := 0$. Prove that $|\cdot|_p$ is an absolute value on $\mathbf{Q}$. That is:

(a) $|x|_p \geq 0$ for all $x \in \mathbf{Q}$, with $|x|_p = 0 \iff x = 0$,

(b) $|xy|_p = |x|_p |y|_p$ for all $x, y \in \mathbf{Q}$,

(c) $|x + y|_p \leq |x|_p + |y|_p$.

Prove that in fact

(c') $|x + y|_p \leq \max\{|x|_p, |y|_p\}$, with equality whenever $|x|_p \neq |y|_p$.

It can be shown that every nontrivial absolute value on $\mathbf{Q}$ is equivalent to either the usual absolute value or one of the p -adic absolute values $|\cdot|_p$ (Ostrowski's theorem).

Problem 7. *partial summation, à la Riemann–Stieltjes* (10 points) Let $x \geq 1$. Let $\{a_n\}$ be a sequence of real numbers, defined for all positive integers $n \leq x$, and let f be a C^1 function on $[1, x]$. For each real t , put $S(t) = \sum_{n \leq t} a_n$. (As in class, sums over integers are assumed to be sums over *positive* integers, and so the condition on n is that $1 \leq n \leq t$.) Show that

$$\begin{aligned} \sum_{n \leq x} a_n f(n) &= S(\lfloor x \rfloor) f(\lfloor x \rfloor) - \int_1^{\lfloor x \rfloor} S(t) f'(t) dt \\ &= S(x) f(x) - \int_1^x S(t) f'(t) dt. \end{aligned}$$

As an application, show that

$$\sum_{p \leq x} \frac{1}{p} = \frac{\pi(x)}{x} + \int_2^x \frac{\pi(t)}{t^2} dt.$$

Problem 8. (5 points) Recall that $T(x) = \log[x]!$. Prove that for some positive constant K and all $x \geq 2$,

$$T(x) \geq x \log x - x - K \log x.$$

I suggest either applying the proof of the integral test or invoking the partial summation formula you just established.